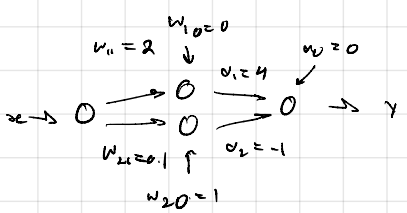
ELEC 5404 Assignment 2 – NN Training

hidden Layer → sigmoid → $\sigma = \frac{1}{1+e^{-r}}$
output → Linear

Q1 Train using Batch Back prop. $\eta = 0.1$, no momentum
calc. new weights after 1 epoch.




Training Data

| i | $x_i$ | $d_i$ |
|---|---|---|
| 1 | 0 | 2 |
| 2 | 1 | 4 |

$$E(w) = \frac{1}{2}\sum (y(x_i, w) - d_i)^2$$

Begin with Forward pass of both samples

$z_1 = \sigma(w_{10} + x w_{11})$ $\quad y = v_0 + v_1 z_1 + v_2 z_2$

$z_2 = \sigma(w_{20} + x w_{21})$

Sample 1

$x_i = 0$

$z_1 = \sigma(0 + 0.2)$
$= \sigma(0)$
$= 0.5$

$z_2 = \sigma(1 + 0 \cdot 0.1)$
$= \sigma(1)$
$= 0.7311$

$y = 0 + 4(0.5) + (-1)(0.7311)$
$= 1.2689$

$e_1 = y - d$
$= 1.2689 - 2$
$= -0.7311$

Sample 2

$x_2 = 1$

$z_1 = \sigma(0 + 1 \cdot 2)$
$= \sigma(2)$
$= 0.8808$

$z_2 = \sigma(1 + 1 \cdot 0.1)$
$= \sigma(1.1)$
$= 0.7503$

$y = 0 + 4(0.8808) + (-1)(0.7503)$
$= 2.7729$

$e_2 = y - d$
$= 2.7729 - 4$
$= -1.2271$

Then Training Error

$E = \frac{1}{2}(e_1^2 + e_2^2)$

$= \frac{1}{2}(0.7311^2 + 1.2271^2)$

$= 1.0201$

Now Derivs. w.r.t each weight

$\frac{\partial y}{\partial w}$ → output Layer

Hidden Layer weights

$z_1 = v_{11} x + v_{10}$

$\frac{dr_1}{dw_{11}} = x$

$\frac{dy}{dw_{11}} = \frac{dy}{dz_1} \cdot \frac{dz_1}{dr_1} \cdot \frac{dr_1}{dw_{11}} \cdot \frac{dr_1}{dw_{10}} = 1$

$= v_1 \cdot z_1(1-z_1) \cdot x$

$\frac{dy}{dw_{10}} = v_1 \cdot z_1(1-z_1) \cdot 1$

$y = v_0 + v_1 z_1 + v_2 z_2$

$\frac{dy}{dv_0} = 1$ $\quad \frac{dy}{dv_1} = z_1$ $\quad \frac{dy}{dv_2} = z_2$

$\frac{dy}{dw_{21}} = \frac{dy}{dz_2} \cdot \frac{dz_2}{dr_2} \cdot \frac{dr_2}{dw_{21}}$

$= v_2 \cdot z_2(1-z_2) \cdot x$ $\quad \frac{dy}{dw_{20}} = v_2 \cdot z_2(1-z_2)$

| weight | $\partial E_i / \partial w$ |
|---|---|
| $v_0$ | $e$ |
| $v_1$ | $e z_1$ |
| $v_2$ | $e z_2$ |
| $w_{10}$ | $e v_1 \cdot z_1(1-z_1)$ |
| $w_{11}$ | $e v_1 \cdot z_1(1-z_1) \cdot x$ |
| $w_{20}$ | $e v_2 \cdot z_2(1-z_2)$ |
| $w_{21}$ | $e v_2 \cdot z_2(1-z_2) \cdot x$ |

Sample 1

$\frac{dE}{dv_0} = -0.7311$

$\frac{dE}{dv_1} = -0.3656$

$\frac{dE}{dv_2} = -0.5341$

$\frac{dE}{dw_{10}} = -0.7311$

$\frac{dE}{dw_{11}} = 0$

$\frac{dE}{dw_{20}} = 0.1437$

$\frac{dE}{dw_{21}} = 0$

Sample 2

$\frac{dE}{dw} = -1.2271$

$\frac{dE}{dv_1} = -1.0808$

$\frac{dE}{dv_2} = -0.9206$

$\frac{dE}{dw_{10}} = -0.5153$

$\frac{dE}{dw_{11}} = -0.5153$

$\frac{dE}{dw_{20}} = 0.2299$

$\frac{dE}{dw_{21}} = 0.2299$

Sum partial derivs. of both samples AND update

$w_{new} = w_{old} - \eta \cdot \frac{dE}{dw}$, with $\eta = 0.1$

| weight | Sample 1 | Sample 2 | dE/dw | $w_{new}$ |
|---|---|---|---|---|
| $v_0$ | -0.7311 | -1.2271 | -1.9581 | 0.1958 |
| $v_1$ | -0.3656 | -1.0808 | -1.4464 | 4.1496 |
| $v_2$ | -0.5341 | -0.9206 | -1.4547 | -0.0545 |
| $w_{10}$ | -0.7311 | -0.5153 | -1.2464 | 0.1246 |
| $w_{11}$ | 0 | -0.5153 | -0.5153 | 2.0515 |
| $w_{20}$ | 0.1437 | 0.2299 | 0.3736 | 0.9626 |
| $w_{21}$ | 0 | 0.2299 | 0.2299 | 0.0440 |